

Square Formula Sheet — Format

canonical layout for `doc/formulas/square/mode_*.tex` (example: `mode_1.tex`)

1. Title **Mode** N --- **Square**; subtitle = the DP granularity flow, coarsest to finest, ending in *add-and-square*.
2. The top DP operation stays on the **left** of every step; refine only the **right** side, one granularity per step, down to DP8 (4×4).
3. Step titles carry **no** parenthetical note — *except* the add-and-square step (keeps the substitution note).
4. Sums in **display** style (limits stacked); **underbrace** each DP-level block with `\DPop{len}{wA}{wB}`.
5. Add-and-square step: **fully expanded**, top DP on the left, the $\frac{1}{2}$ **inside** each underbrace (so the braced group is one DP8 (4×4)), the 2^k weight **outside**.
6. Final step *All components*: $\boxed{\text{PE} / \alpha / \beta}$ — PE = squares, α = A-only, β = B-only; one term per DP8 (4×4), same order in every row.
7. Keep to one page where possible: tune the `\[. .]` gaps and the margin.

The reusable preamble and a copy-ready skeleton are in the comment block at the top of this file.